25900084 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCFTQ100 - FOOD ADDITIVES

(2024 ADMISSION ONWARDS)

Duration: 1 Hours **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **5** questions

Each question carries **1** marks

1. Which method is used to extract colorants from citrus fruits? [K] / [CO5]
2. Name 2 sequestrants. [K] / [CO4]
3. Give two examples of polyols. [K] / [CO5]
4. What is one carrier material used in the microencapsulation process? [U] / [CO5]
5. Name one example of a natural hydrocolloid derived from seaweed. [K] / [CO5]
6. What are the uses of corn sweeteners in food production? [K] / [CO5]
7. Name one common antifoaming agent used in food processing. [K] / [CO3]

[1x5 = 5]

Part B

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

8. Classify food additives based on their function and provide examples for each category. [K] / [CO1]

9. How do preservatives help prevent oxidation in food? [K] / [CO3]

10. What do you understand by antimicrobial agents? Provide examples. [K] / [CO2]

11. What are the benefits of using honey in recipes instead of sugar? [K] / [CO5]

12. How do buffering salts help maintain flavor and prevent extreme pH changes in food? [K] / [CO5]

13. What are the uses of flavorings and sweeteners in the food industry. [K] / [CO5]

14. How does citric acid help preserve the color of fruits and vegetables? Provide examples of its application in food. [K] / [CO5]

[4x5 = 20]

Part C

Long Essay Questions

Answer any **1** questions

Each question carries **10** marks

15. Discuss the role of acidulants and anticaking agents in food production? [U] / [CO5]

16. How do leavening agents contribute to the baking process? Discuss their impact on the texture and volume of baked goods. [K] / [CO5]

17. What is the mechanism by which antioxidants prevent the oxidation of food components? Discuss how they contribute to maintaining food quality during storage. [K] / [CO5]

[10x1 = 10]